ELEC3104: Digital Signal Processing

Topic 1: DSP fundamentals

- Answers to Multiple Choice Questions

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Topic 1: Tutorial Questions (SOLO - Level 1)

1. An analogue signal contains frequencies up to 10 kHz. Suppose that we sample this signal at a sampling frequency f_s = 8kHz. Examine what will happen to the frequency f_1 = 5 kHz? (Tick the correct answer)

   a) 5kHz frequency will remain unchanged
   ✓b) 5kHz will alias to 3kHz
   c) 5kHz will alias to 0Hz
   d) None of the above

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f_1 (5 kHz) > $\frac{f_s}{2}$ ($\frac{8\,kHz}{2}$) ∴ Aliasing will occur

New frequency is $f_1' = f_1 - k f_s = 5000 - 1 \times 8000$ (↑ $k=1$) $= -3000$ Hz

i.e 5 kHz will alias to 3 kHz

2. Which of the following is a not periodic signal ? (Tick the correct answer)

   a) x[n] = sin[πn/9] ← Period = 18
   b) x[n] = sin[nπ] ← Period = 2
   ✓c) x[n] = cos[πn²/15] ← non Periodic (because of n^2)
   d) x[n] = sin[πn/5 + π] + cos[πn/10 − π] ← Period = least common multiple of N_1 & N_2 = LCM(10, 20) = 20

   Period N_1 = 10; Period N_2 = 20

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3. Given a real-valued bandpass signal x_a(t) with a bandpass spectrum of X_a(f) nonzero only for $9\ kHz < |f| < 12kHz$, determine the minimum sampling frequency (f_s) for which no aliasing occurs? (Tick the correct answer)

   a) f_S = 18 kHz
   ✓b) f_S = 6 kHz
   c) f_S = 24 kHz
   d) None of the above

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Bandwidth (B) = 12 − 9 kHz = 3 kHz

f_h = 12 kHz which is integer multiple of the Bandwidth.

∴ f_s (minimum) = 2 × B = 2 × 3 kHz = 6 kH

4. A discrete-time signal x(n) = 6.35 cos(nπ/10) is quantised with a resolution Δv=0.2 $volts$. How many bits are required in the Analogue to Digital Converter? (Tick the correct answer)

   ✓a) 6 bits
   b) 8 bits
   c) 12 bits

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$\Delta V = \frac{2 \times 6.35}{2^B - 1}$ (↑ 0.2V)

B - number of bits

$2^B - 1 = 63.5$

∴ B = 6

Topic 1: Tutorial Questions (SOLO - Level 1)

5. Consider the amplitude modulated signal $x(t) = 5\,cos(3\pi t)\,cos(\pi t)$. Determine its spectral frequency components [you may assume that the frequencies are in kHz) (Tick the correct answer)
   a) $\pm$1.5kHz and $\pm$0.5kHz
   b) $\pm$3kHz and $\pm$1kHz
   c) $\pm$2kHz
   ✓d) $\pm$2kHz and $\pm$1kHz

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$x(t) = 5\cos(2\times1.5\pi t)\cdot\cos(2\times0.5\pi t)$

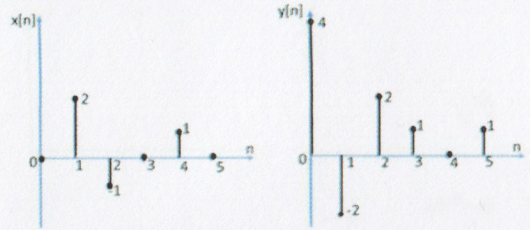
f_1 f_2

Product of two real sinusoids results in the sum and difference frequencies.

ie $\pm(f_1+f_2)$ & $\pm(f_1-f_2)$

$\pm$ 2kHz & $\pm$ 1kHz

6. Given the discrete-time sequences x[n] and y[n] shown, which difference formula best describes their relationship ? (Tick the correct answer)



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$x[0]=0$; $y[0]\neq0$ $(y[0]=4)$

$n=0$ $n=0$

∴ The signal x[n] must have been advanced by one sample

   ✓a) y[n] = x[n-1] + 2x[n+1]
   b) y[n] = 2x[n] – x[n-1]
   c) y[n] = 2x[n-1] + x[n-2]
   d) y[n] = 2x[n-2] – x[n]

7. Determine the period of the discrete time signal $x[n] = \sin(0.2n + \pi)$. (Tick the correct answer)
   (a) 1 sample (b) 5 samples
   (c) π samples ✓(d) Non of the above

Topic 1 : Page 8 Example 4

Topic 1: Tutorial Questions (SOLO-Level 2)

1. A sinusoidal signal with peak-to-peak amplitude of 5V is sampled at 50kHz with uniform quantisation. Find the minimum number of bits for the analogue to digital converter to achieve a SQNR of at least 92dB. State any assumptions made.

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$SQNR = 6.02B + 1.76\,dB$ (B-number of bits)

$SQNR \geq 92\,dB$

$92 \leq 6.02B + 1.76$

$\therefore B \geq 14.9$

$\therefore \boxed{B = 15}$ (minimum number of bits)

2. Show that the signal to quantisation noise ratio (SQNR) of a linear B-bit analogue-to-digital converter is given by

$$SQNR = 6.02B + 4.77 - 20\log\left(A/\sigma_{sig}\right)$$

where the input range of the A/D converter is $\pm$A and the rms value of the input signal is σ_{sig}. Determine the SQNR if B is 16 bits and the input is a signal with a rms value of A/5.

Topic 1: Page 29 → $SQNR = 10\log\frac{\sigma_s^2}{\sigma_e^2} = 10\log\frac{\sigma_s^2}{(\Delta V^2/12)} = 10\log\frac{\sigma_s^2}{(2A/2^B)^2/12} = 10\log\sigma_s^2 \cdot \frac{2^{2B} \times 3}{A^2}$

$\therefore SQNR = 10\log\frac{\sigma_s^2}{A^2} + 10\log 2^{2B} + 10\log 3$

$= 20\log\frac{\sigma_s}{A} + B(10\log 2^2) + 4.77$

$= 6.02$

$\boxed{SQNR = 6.02B + 4.77 - 20\log\frac{A}{\sigma_s}}$

$\sigma_s = \sigma_{sig}$

$SQNR = 6.02 \times 16 + 4.77 - 20\log\frac{A}{(A/5)} = 6.02 \times 16 + 4.77 - 20\log(5)$

$\boxed{SQNR = 87.1\,dB}$

Topic 1: Tutorial Questions (SOLO-Level 2)

3. An analogue signal $x_a(t) = \cos(2\pi 500t)$ is sampled at a rate of f_s=4kHz. Determine the resulting analogue and digital magnitude spectra.

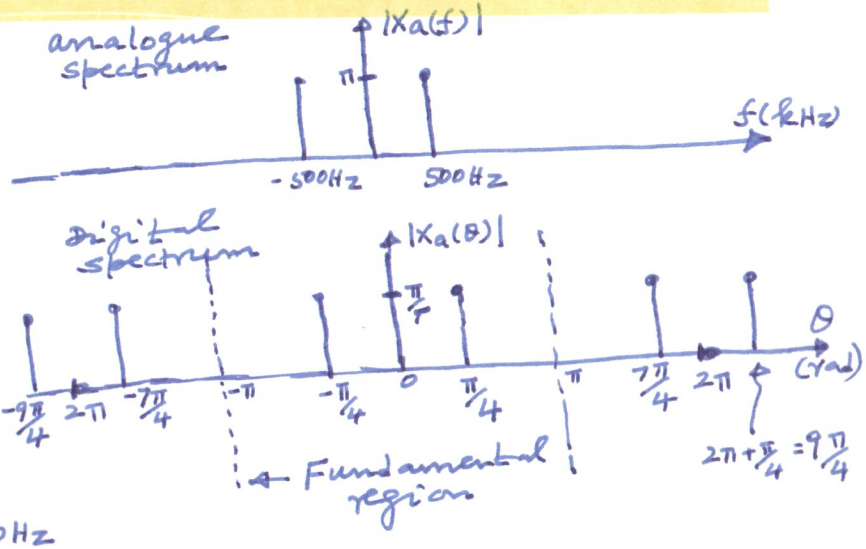
$FT[\cos\omega_1 t] = \pi\delta(\omega-\omega_1) + \pi\delta(\omega+\omega_1)$

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$x_a(t) = \cos(2\pi 500 t)$

$\therefore f_1 = 500\,Hz$

$\therefore$ Corresponding digital frequency $\theta = 2\pi\frac{f_1}{f_s} = 2\pi\times\frac{500}{4000} = \frac{\pi}{4}$



4. An analogue signal $x(t) = \sin(480\pi t) + 3\sin(720\pi t)$ is sampled 600 times per sec:

(a) Determine the Nyquist sampling rate for x(t).

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$x(t) = \sin(2\pi\times240t) + 3\sin(2\pi\times360t)$

$f_1 = 240\,Hz$; $f_2 = 360\,Hz \Rightarrow f_{max} = 360\,Hz \therefore F_{Nyquist} = 2\times f_{max} = 720\,Hz$

(b) Determine the folding frequency (or half the sampling frequency).

$f_s = 600\,Hz \therefore f_{fold}$ or $\frac{f_s}{2} = 300\,Hz$

(c) What are the frequencies, in radians, in the resulting discrete time signal x[n]?

$f_2 > \frac{f_s}{2}$ ($\therefore$ it is affected by aliasing) $\therefore$ new frequency due to aliasing $f_2' = f_2 - k\cdot f_s = 360 - 1\cdot 600 = -240\,Hz$ ($k=1$)

(d) If x[n] is passed through an ideal D/A converter what is the reconstructed signal y(t)?

We have $t = nT = \frac{n}{f_s} = \frac{n}{600}$

$y(t) = -2\sin(2\pi 240 t)$

$\therefore y(t) = -2\sin(480\pi t)$

$\therefore x[n] = \sin\left(2\pi\frac{240}{600}n\right) + 3\sin\left(2\pi\frac{(-240)}{600}n\right)$

$x[n] = -2\sin\left(\frac{4\pi}{5}n\right)$

$x[n] = -\sin\left(2\pi\frac{240}{600}n\right)$

note: $\frac{n}{600} = t$

Topic 1: Tutorial Questions (SOLO-Level 2)

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5. A signal of $v(t) = 4.5\sin(1000\pi t)$ Volts is sampled at 2.5 kHz. An eight-bit A/D converter with a nominal input range of ± 5 Volts is used to convert sampled voltages to integer codes.

(a) Calculate the quantization interval of the A/D converter.

(b) Determine the maximum quantization error.

(a) Peak to Peak amplitude = 10 V

Quantisation interval or resolution (ΔV) = $\frac{10V}{2^B - 1} \approx \frac{10V}{2^B}$ (if B is Large)

$\therefore \Delta V = \frac{10V}{2^8} = 39.062\ mV$

↑ 8-bit A/D converter

(b) Maximum Quantisation error $e = \frac{\Delta V}{2} = 19.53\ mV$

Topic 1: Tutorial Questions (SOLO-Level 2)

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6. Consider the discrete-time sequence $x[n] = \cos[(n\pi/8) + (\pi/4)]$.
   (i) Determine the fundamental period of x[n].
   (ii) If the sampling frequency f_s = 10 kHz, write the equation of the analogue signal x(t).

(i) $x[n] = \cos[n\theta + \pi/4]$

digital frequency $\theta = \frac{\pi}{8}$; The fundamental period $N = \frac{2\pi k}{\theta}$

$N = \frac{2\pi \times 1}{\pi/8}$ ($k=1$) $= 16$ samples

Period = 16 samples

(ii) $\theta = \frac{\pi}{8}$; $2\pi \frac{f_a}{f_s} = \frac{\pi}{8} \Rightarrow \frac{f_a}{f_s} = \frac{1}{16} \Rightarrow f_a = \frac{10000}{16}$

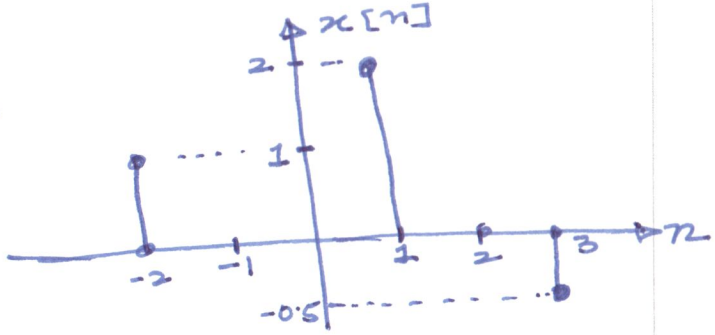
$= 625$ Hz

$x[n] = \cos(n\theta + \pi/4)$

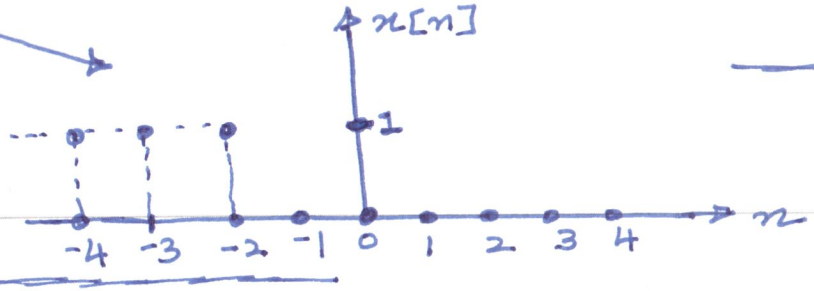
$x(t) = \cos(2\pi(625)t + \pi/4)$

7. Sketch the following:
   (c) $x[n] = \delta[n+2] + 2\,\delta[n-1] - 0.5\,\delta[n-3]$

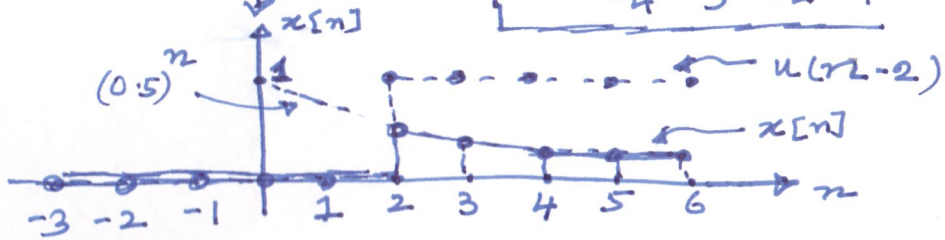
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   (d) $x[n] = u[-n-2]$



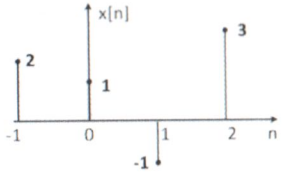
   (e) $x[n] = 0.5^n\, u[n-2]$



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8. Write the equation for x[n] (see diagram) as a sum of scaled and delayed unit impulses

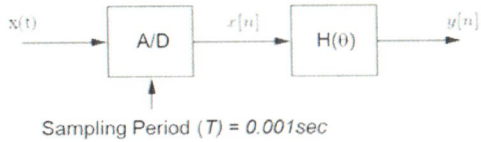


$x[n] = 2\delta(n+1) + \delta(n) - \delta(n-1) + 3\delta(n-2)$

9. For the system shown below , the input x(t) = cos(400πt) + cos(600πt). Find the output y[n] where

$f_1 = 200$ Hz, $f_2 = 300$ Hz

$$H(\theta) = \begin{cases} 1 & |\theta \le 0.5\pi| \\ 0 & otherwise \end{cases}$$



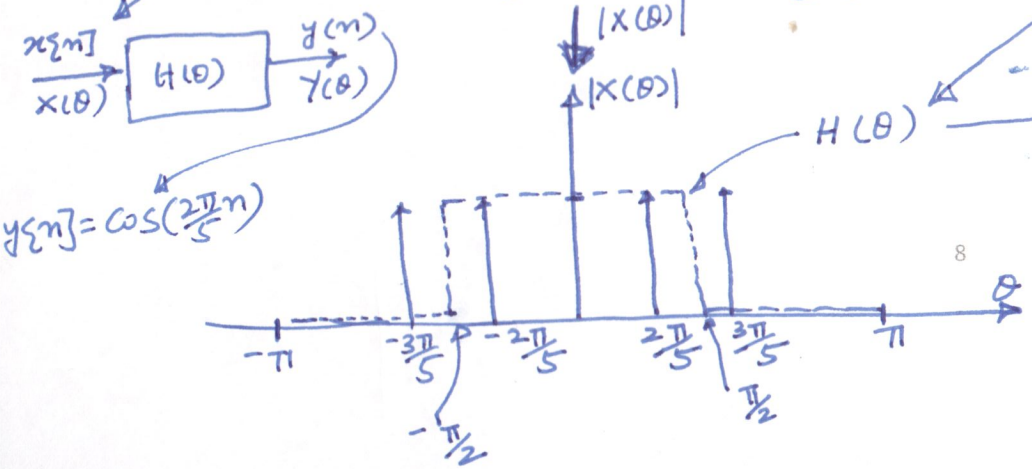
$f_s = \frac{1}{0.001} = 1000$ Hz

Sampling frequency

x(t) to x[n] transformation

$x[n] = \cos\left(2\pi \frac{200}{1000} n\right) + \cos\left(2\pi \frac{300}{1000} n\right)$

$x[n] = \cos\left(\frac{2\pi}{5} n\right) + \cos\left(\frac{3\pi}{5} n\right)$



$y[n] = \cos\left(\frac{2\pi}{5} n\right)$

The filter BW is $\pi/2 = 0.5\pi$ it removes the two frequencies $\pm \frac{3\pi}{5}$ and hence the term $\cos(\frac{3\pi}{5})$ is removed.

$\therefore \boxed{y[n] = \cos\left(\frac{2\pi}{5} n\right)}$

Topic 1: Tutorial questions (SOLO - Level 3)

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1. A signal of $v(t) = 4.5\sin(1000\pi t)$ Volts is sampled at 2.5 kHz. An eight-bit A/D converter with a nominal input range of ±5 Volts is used to convert sampled voltages to integer codes. At what time (t>0) does the quantization error first reach its maximum value?

$\Delta V = \frac{10}{2^8} = 39.06\,mV$; Maximum quantisation error (e_m) = $\frac{\Delta V}{2}$

We will denote at $t = t_1$, the maximum quantisation error occurs

$\therefore \frac{\Delta V}{2} = 4.5\sin(1000\pi t_1) \Rightarrow t_1 = \left\{\sin^{-1}\left[\frac{\Delta V}{9}\right]\right\} \Big/ (1000\pi)$, $\therefore$ $t_1 = 1.38\,\mu s$

Note: $t_1 = 1.38\,\mu s$ suggests that the e_m will oscillate at a frequency ($\approx 362\,kHz$) much higher than the sampling frequency of 2.5 kHz ; $\therefore$ Quantization error appears as noise at the output.

2. A system samples a signal $x(t) = \cos(\omega_o t)\cos(3\omega_o t)$ at f_s = 7 kHz and reconstructs the signal using an ideal LPF with a cutoff frequency of 3.5 kHz. The value of ω_o is $2\pi \times 10^3$ rad.

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(a) What is the highest frequency component present in the signal?

Hint : $\cos(\alpha)\cos(\beta) = \frac{1}{2}[\cos(\alpha+\beta) + \cos(\alpha-\beta)]$

(b) Sketch |X(ω)| and |X(θ)| – the magnitude spectrum of the signal before and after sampling for (a) above.

(c) Can aliasing be prevented if the sampling frequency is increased to 8kHz? Explain.

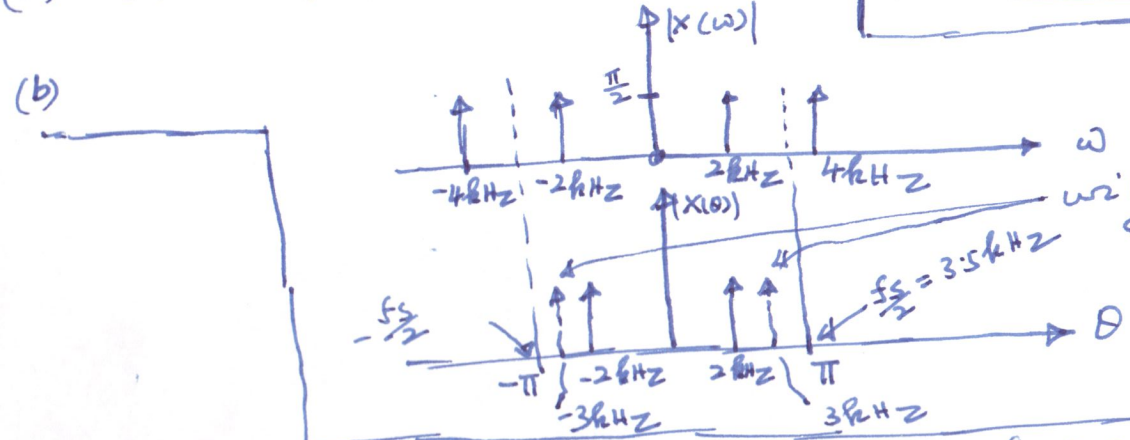
(a) $x(t) = \cos(\omega_o t)\cdot\cos(3\omega_o t) = \frac{1}{2}[\cos(4\omega_o t) + \cos(2\omega_o t)]$

Highest frequency component

$\omega = 4\omega_o = 4 \times 2\pi \times 10^3$

$\therefore$ 4 kHz

(b)



$f_s = 7\,kHz$ $\frac{f_s}{2} = 3.5\,kHz$

4 kHz will be aliased.

$\therefore$ New frequency $f_2' = f_2 - k \times f_s$

$= 4 - 1.7$

$= -3\,kHz$

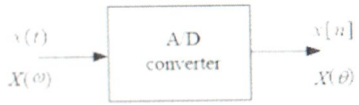
(c) NO. To avoid aliasing, we need $f_s > 2\,f_{highest}$ ($f_{highest} = 4\,kHz$)

$\therefore$ $f_s > 8\,kHz$

Topic 1: Tutorial questions (SOLO - Level 3)

3. If an analogue signal $x(t)$ is sampled every T seconds, then at the output of the Analogue to Digital Converter, as shown in a diagram below. Using Inverse Fourier Transforms in Continuous time and discrete time domains, show that

$$X(\theta) = \frac{1}{T}\sum_{k=-\infty}^{\infty} X(\omega + k\frac{2\pi}{T})$$



IFT (Inverse Fourier Transform) in continuous time is given by

$$x(t) = \frac{1}{2\pi}\int_{-\infty}^{\infty} X(\omega)\, e^{j\omega t}\, d\omega$$

given $x[n] = x(t)\Big|_{t=nT}$

$$x[n] = \frac{1}{2\pi}\int_{-\infty}^{\infty} X(\omega)\, e^{j\omega(nT)}\, d\omega$$

$e^{j\omega nT}$ is Periodic function of period 2π

$$\therefore x[n] = \frac{1}{2\pi}\sum_{k=-\infty}^{\infty}\int_{2k\pi-\pi}^{2k\pi+\pi} X(\omega)\, e^{j\omega nT}\, d(\omega T)\times\frac{1}{T}$$

$$= \frac{1}{2\pi}\sum_{k=-\infty}^{\infty}\frac{1}{T}\int_{-\pi}^{\pi} X(\omega + k\tfrac{2\pi}{T})\, e^{j(\omega+k\frac{2\pi}{T})nT}\, d(\omega T)$$

$$= \frac{1}{2\pi}\cdot\frac{1}{T}\sum_{k=-\infty}^{\infty}\int_{-\pi}^{\pi} X(\omega + k\tfrac{2\pi}{T})\, e^{j\omega nT}\, d(\omega T)$$

let $\omega T = \theta$

$$x[n] = \frac{1}{2\pi}\int_{-\pi}^{\pi}\underbrace{\frac{1}{T}\sum_{k=-\infty}^{\infty} X(\omega + k\tfrac{2\pi}{T})}_{=X(\theta)}\, e^{j\theta n}\cdot d\theta$$

$$\therefore X(\theta) = \frac{1}{T}\sum_{k=-\infty}^{\infty} X(\omega + k\tfrac{2\pi}{T})$$

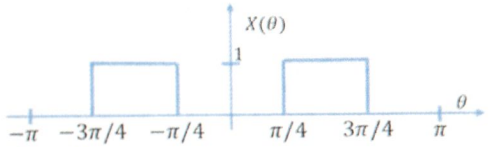
IDFT (for Discrete signals)

$$x[n] = \frac{1}{2\pi}\int_{-\pi}^{\pi} X(\theta)\, e^{j n\theta}\, d\theta$$

Topic 1: Tutorial questions (SOLO - Levels 3)

4. Find the Inverse DTFT of $X(\theta)$ for the figure shown below and show that $x[n] = \frac{2\sin(n\pi/4)\cos(n\pi/2)}{n\pi}$

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$$X(\theta) = \sum_{n=-\infty}^{\infty} x(n)e^{-jn\theta}$$

DTFT pair

$$x(n) = \frac{1}{2\pi}\int_{-\pi}^{\pi} X(\theta)e^{jn\theta}\,d\theta$$

θ = digital frequency; $-\pi \le \theta \le \pi$

Using the inverse DTFT, we have

$$x[n] = \frac{1}{2\pi}\int_{-\pi}^{\pi} X(\theta)e^{jn\theta}\,d\theta = \frac{1}{2\pi}\int_{\pi/4}^{3\pi/4} 1\cdot e^{jn\theta}\,d\theta + \frac{1}{2\pi}\int_{-3\pi/4}^{-\pi/4} 1\cdot e^{jn\theta}\,d\theta$$

$$= \frac{1}{2\pi jn}\left[e^{jn\theta}\right]_{\pi/4}^{3\pi/4} + \frac{1}{2\pi jn}\left[e^{jn\theta}\right]_{-3\pi/4}^{-\pi/4}$$

$$x[n] = \frac{1}{2\pi jn}\left[e^{j\frac{3\pi}{4}n} - e^{j\frac{\pi}{4}n}\right] + \frac{1}{2\pi jn}\left[e^{-j\frac{\pi}{4}n} - e^{-j\frac{3\pi}{4}n}\right]$$

Rearranging the terms, we have

$$x[n] = \frac{1}{2\pi jn}\left[e^{j\frac{3\pi}{4}n} - e^{-j\frac{3\pi}{4}n}\right] - \frac{1}{2\pi jn}\left[e^{jn\frac{\pi}{4}} - e^{-jn\frac{\pi}{4}}\right]$$

$$= \frac{1}{\pi n}\left[\frac{e^{j\frac{3\pi}{4}n} - e^{-j\frac{3\pi}{4}n}}{2j}\right] - \frac{1}{\pi n}\left[\frac{e^{jn\frac{\pi}{4}} - e^{-jn\frac{\pi}{4}}}{2j}\right]$$

$$x[n] = \frac{\sin\left(\frac{3\pi}{4}n\right)}{\pi n} - \frac{\sin\left(\frac{\pi}{4}n\right)}{\pi n}$$

$$\boxed{x[n] = 2\,\frac{\sin\left(\frac{n\pi}{4}\right)\cdot\cos\left(\frac{n\pi}{2}\right)}{n\pi}}$$

$$\left\{\sin A - \sin B = 2\cos\frac{A+B}{2}\cdot\sin\frac{A-B}{2}\right\}$$

Topic 1: Tutorial questions (SOLO - Level 3)

5. Find the Fourier Transform of x(t) = δ(t-a). Hence, show that $\int_{-\infty}^{\infty} e^{j\omega t}\,dt = 2\pi\delta(\omega)$.

$$X(\omega) = \int_{-\infty}^{\infty} x(t)\, e^{-j\omega t}\, dt$$

$\delta(t-a) = x(t)$

$$= \int_{-\infty}^{\infty} \delta(t-a)\, e^{-j\omega t}\, dt$$

$\delta(t-a) = 1$ at $t = a$

$$\boxed{X(\omega) = e^{-j\omega a}}$$

$$FT\{\delta(t-a)\} = e^{-j\omega a}$$

$$\therefore FT^{-1}\{e^{-j\omega a}\} = \delta(t-a)$$

(Inverse)

$$\frac{1}{2\pi}\int_{-\infty}^{\infty} e^{-j\omega a}\cdot e^{j\omega t}\, d\omega = \delta(t-a)$$

Inverse Fourier Transform

$$x(t) = \frac{1}{2\pi}\int_{-\infty}^{\infty} X(\omega)\, e^{j\omega t}\, d\omega$$

$$\int_{-\infty}^{\infty} e^{j\omega(t-a)}\, d\omega = 2\pi\delta(t-a)$$

If $a = 0$

$$\int_{-\infty}^{\infty} e^{j\omega t}\, d\omega = 2\pi\delta(t)$$

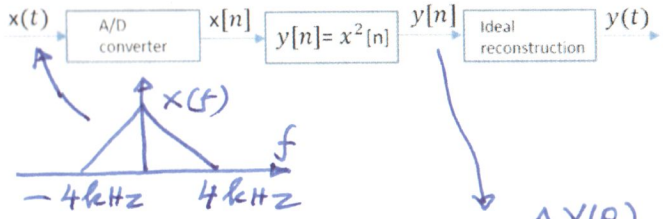
Interchanging the variables, we also have

$$\boxed{\int_{-\infty}^{\infty} e^{j\omega t}\, dt = 2\pi\delta(\omega)}$$

i.e. $t = \omega$ $\therefore dt = d\omega$ & the limits will be the same

Topic 1: Tutorial questions (SOLO - Level 4)

1. A speech signal is band limited to 4 kHz is ideally sampled at a sampling rate, f_s, and the sampled signal $x[n]$, is processed by squaring operations $y[n] = x^2[n]$, whose output is ideally reconstructed to obtain y(t) (see diagram on the right):



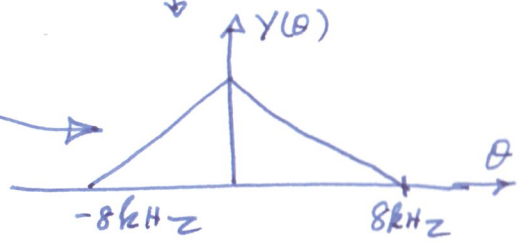
What minimum sampling rate f_s, will ensure $y(t) = x^2(t)$?

let $x(t) = A\sin(\omega_0 t)$

$y(t) = [x(t)]^2 = A^2\sin^2(\omega_0 t) = \frac{A^2}{2}(1 - \cos 2\omega_0 t)$

(squaring operation produces $2\omega_0$)

∴ The spectrum of $y[n]$ extends to twice the frequency of $x[n]$



∴ minimum sampling frequency should be 16kHz

2. The signal $x(t) = 2\cos(2000\pi t) - 4\sin(4000\pi t)$ is quantised by rounding, using a 12 bit quantiser. What is the rms quantisation error and the SQNR?

The signal Power $(P_s) = \left(\frac{2}{\sqrt{2}}\right)^2 + \left(\frac{4}{\sqrt{2}}\right)^2 = \sqrt{5}$

The peak value of $x(t)$ cannot exceed 6, so the peak to peak amplitude is ≈ 12.

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∴ $\Delta V = \frac{12}{2^B - 1} = \frac{12}{2^{12} - 1}$ (using 12 bit quantiser) $\approx \frac{12}{2^{12}}$

RMS quantisation error $= \frac{\Delta V}{\sqrt{12}} = \frac{12/2^{12}}{\sqrt{12}} = 0.0008457$

$SQNR\,(dB) = 10\,Log\,\frac{P_s(\text{signal Power})}{P_n(\text{noise Power})} = 10\,Log\,\frac{\sqrt{5}}{\frac{\Delta V^2}{12}} = 10\,Log\,\frac{\sqrt{5}}{(12/2^{12})^2/12}$

$= 64.95$

Topic 1: Tutorial questions (SOLO-Level 5)

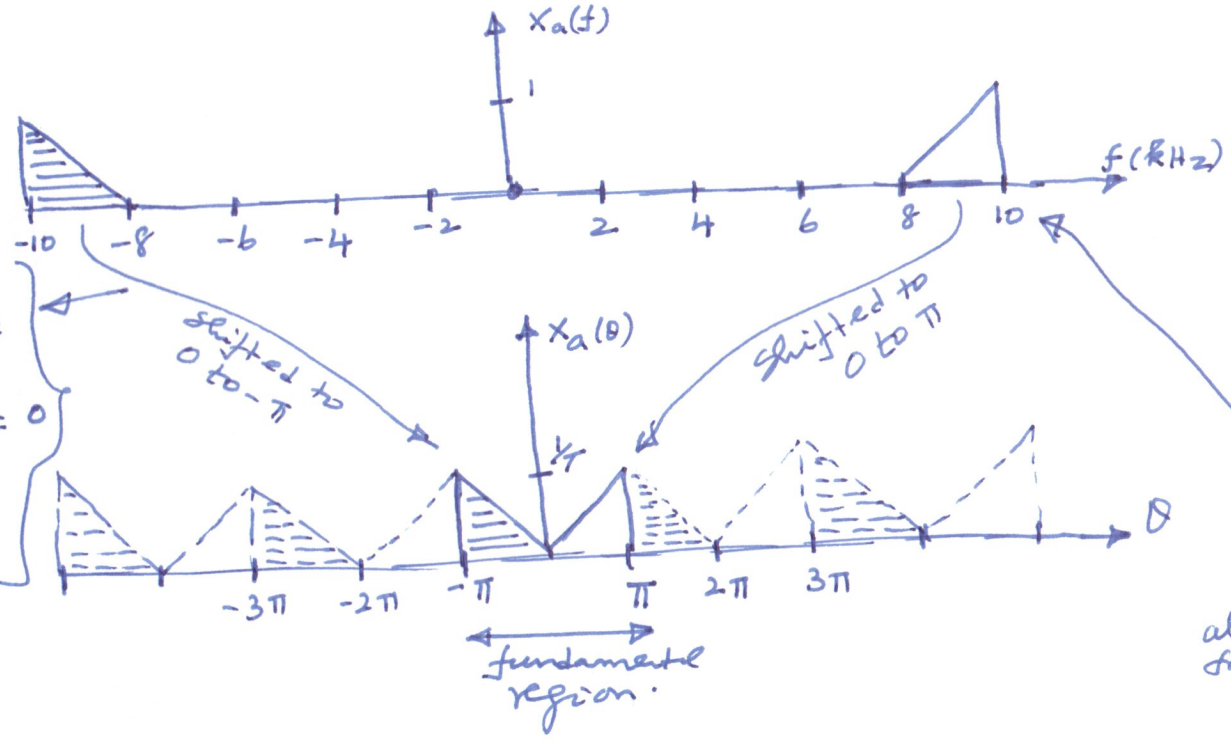
1. Given a real-valued bandpass signal $x_a(t)$ and $X_a(f) = 0$ for $|f| < f_1$ and $|f| > f_2$, the Nyquist sampling theorem says that the minimum sampling frequency is $f_s = 2f_2$. However , in some cases , the signal may be sampled at a lower rate.

   (a) Suppose that f_1 = 8 kHz and f_2 = 10 kHz. Draw a sketch of $X_a(\theta)$, if the sampling frequency f_s = 4 kHz.

Note

$$X(\theta) = \frac{1}{T}\sum_{k=-\infty}^{\infty} X\left(\omega + k\frac{2\pi}{T}\right)$$

$X(\theta)$ is formed by shifting $X_a(f)$ by integer multiples of the sampling frequency.



$f_3 = -8\,kHz$

$f_4 = -10\,kHz$

$f_3' = f_3 - k f_s = -8 - (-2)4 = 0$

$f_4' = f_4 - (-2)4 = -2\,kHz$

$f_s = 4\,kHz$,

$f_1 = 8\,kHz$ & $f_2 = 10\,kHz$

$f_1' = f_1 - k f_s = 8 - 2\times4 = 0$ [$k=2$]

aliased frequency → $f_2' = f_2 - k f_s = 10 - 2\times4 = 2\,kHz$

Topic 1: Tutorial questions (SOLO-Level 5)

1. (b) If the bandwidth of the bandpass signal is defined as $B = f_2 - f_1$ and the centre frequency to be $f_c = \frac{(f_2+f_1)}{2}$, Show that if $f_c > B/2$ and f_2 is an integer multiple of the bandwidth B, no aliasing will occur if $x_a(t)$ is sampled at a sampling frequency $f_s = 2B$.

If f_2 is an integer multiple of B, we may express f_1 & f_2 as follows

$f_1 = (m-1)B$; $f_2 = mB$ m is an integer ; also $f_2 - f_1 = B$

With the sampling frequency $f_s = 2B$, the sampled signal has a spectrum

$$X_a(0) = \frac{1}{T}\sum_{-\infty}^{\infty} X_a(f - k\underbrace{2B}_{f_s})$$

If $X_a(f)$ is non zero only for $(m-1)B < |f| < mB$, there is only one term in the sum that contributes to $X_a(f)$ in the frequency range $0 < f < B$ and only one term that contributes to the frequency range $-B < f < 0$ (draw a sketch similar to Part a). Therefore, there is no aliasing and $x_a(t)$ may be sampled without aliasing if $f_s = 2B$.

1. (c) Repeat part (b) for the case in which f_2 is not an integer multiple of the bandwidth B.

If f_2 is not an integer multiple of B, we may always increase B until this is the case.

let $k = \left[\frac{f_2}{B}\right]$ take only integer part

Now, if we simply increase B to B' where

$k = \frac{f_2}{B'}$ (or $B' = \frac{f_2}{k}$)

and f_2 now becomes integer multiple of B'.

$\therefore x_a(t)$ may be sampled without aliasing, a sampling frequency of

$$f_s = 2B' = \frac{2f_2}{\left[\frac{f_2}{B}\right]_{\text{integer part only}}}$$

Topic 1: Tutorial questions (SOLO-Level 5)

2. The D/A output approximates the analogue signal by a series of rectangular pulses, whose heights are equal to the corresponding value of the signal pulse as shown in Figure A.

a) Consider one pulse, $\hat{y}(t)$, with a height k and show that the corresponding frequency response $\hat{Y}(\omega)$ is a sinc function given by the following equation:

$$\hat{Y}(\omega) = k\,\frac{T}{2}\,e^{-\frac{j\omega T}{2}}\,\frac{\sin\frac{\omega T}{2}}{\frac{\omega T}{2}}$$

b) Explore the effect of $\hat{Y}(\omega)$ on the output signal spectrum $Y(\omega)$ and illustrate your understanding of this using a spectral plot. If compensation is required for the $sinx/x$, roll-off inherent in the D/A process, what function would you use to compensate? Briefly explain why.

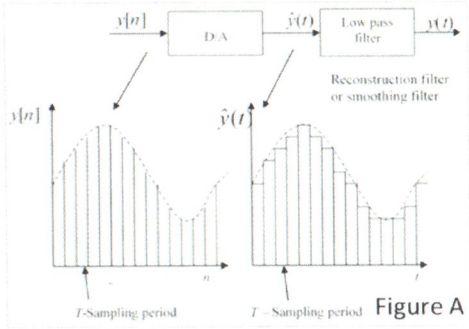
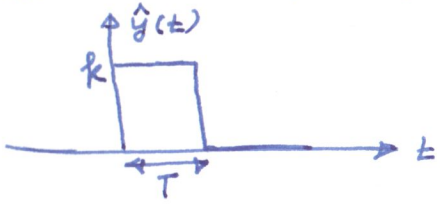


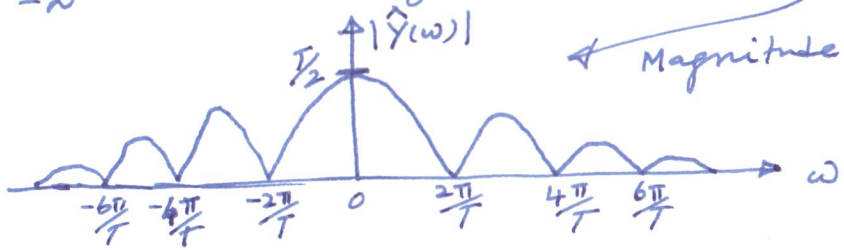
Figure A

(a) Just consider one pulse:



The corresponding frequency response is

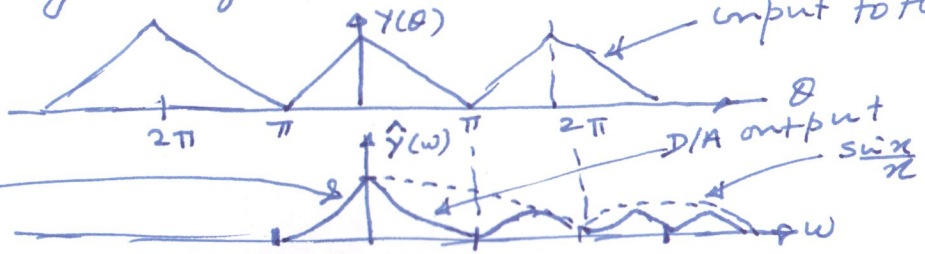
$$\hat{Y}(\omega) = \int_{-\infty}^{\infty} \hat{y}(t)\,e^{-j\omega t}\,dt = \int_{0}^{T} e^{-j\omega t}\,dt = \frac{T}{2}\,e^{\frac{j\omega T}{2}}\,\frac{\sin\frac{\omega T}{2}}{\frac{\omega T}{2}}$$



← Magnitude spectrum $\left(\frac{\sin x}{x}\right)$

(b) In the frequency domain, the staircase of the D/A converter introduces a type of distortion known as $\frac{\sin x}{x}$ where $x = \frac{\omega T}{2}$. The amplitude of the output signal spectrum $Y(\omega)$ is multiplied by the $\frac{\sin x}{x}$ function, which acts like a low pass filter with the high frequency heavily attenuated.

The $\frac{\sin x}{x}$ effect is due to the holding action of the D/A and in signal recovery introduces an amplitude distortion



This distortion can be eliminated by equalisation. Before converting to analogue form the signal is passed through a digital filter whose amplitude response has a $\frac{x}{\sin x}$ shape.